

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. (ME) Examination

November 2013

ME102 Fundamentals of Mechanical Engineering

DATE: 03.12.2013, TUESDAY TIME: 10:00 A.M. TO 01:00 P.M.

MAXIMUM MARKS: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Property table is given at the end of paper.

SECTION-I

Q-1 (a) Explain the following term in brief. (Any Three)

[06]

- | | |
|-------------------------|--------------------------------|
| 1. Specific Heat | 2. Enthalpy |
| 3. Thermodynamic system | 4. First law of thermodynamics |

Q-2 (a) Derive the expression of relation between two specific heats of perfect gases.

[07]

- (b) 0.5 kg of an ideal gas having volume of 0.08 m^3 is expanded at constant pressure of 6 bar in a cylinder until its volume becomes 0.12 m^3 . The gas is then expanded isentropically until its volume becomes 0.30 m^3 . Calculate the temperature and pressure at the end of each process, also calculate total work done. Take $C_p = 0.490 \text{ kJ/kg K}$, $C_v = 0.360 \text{ kJ/kg K}$. [07]

OR

Q-2 (a) Explain construction and working of Cochran boiler with neat sketch.

[07]

- (b) What is boiler mountings and accessories? Explain working of pressure gauge and feed check valve with neat sketch. [07]

Q-3 (a) Define scavenging and Explain the working of four stroke petrol engine with neat sketch.

[08]

- (b) Derive the expression for work for reciprocating air compressor without clearance.

[07]

OR

Q-3 (a) A single cylinder two stroke diesel engine has torque of 650 Nm at speed of 400 rpm. It has bore of 20 cm and stroke length is 30 cm. Oil consumption during the test is 8.5 [07]

kg/hr. Mean effective pressure is recorded as 5.5 bar. The fuel has calorific value of 42.5 MPa. Calculate (i) Mechanical Efficiency (ii) indicated thermal efficiency (iii) brake thermal efficiency (iv) specific fuel consumption in kg/kWh.

(b) Explain the following in brief. (Any Four) [08]

1. Oldham coupling.
2. Centrifugal clutch.
3. Reverted Gear Train.
4. Use of compressed air.
5. Draw PV diagram of Otto cycle.
6. Helical gear.

SECTION – II

Q- 4(a) Explain the steam formation with T-h diagram. [07]

(b) Enlist required properties of good refrigerant. [02]

Q-5 (a) Classify the fuels. Explain Bio-fuels in detail. Give advantages and limitation of bio-fuels. [07]

(b) Determine enthalpy and internal energy of 1 kg of steam at a pressure 12 bar when (i) [05]
the dryness fraction of steam is 0.8 (ii) steam is dry and saturated (iii) steam is
superheated to 280°C. take $C_{ps} = 2.1 \text{ kJ/kg K}$.

OR

Q-5 (a) Explain the Vapour Compression Refrigeration system with neat line diagram. [07]

(b) Explain Air Chamber in pump with neat sketch. [05]

Q-6(a) What is priming in pump? Explain various methods for priming. [07]

(b) Give comparison between Belt drive, Chain drive, and Gear drive. [07]

OR

Q-6 (a) Explain Multi staging of reciprocating air compressor in detail. [07]

(b) Enlist types of centrifugal pump and explain construction and working of diffuser type centrifugal pump. [07]
